

Companion XLII: The Kinematic Sunyaev–Zel’dovich Effect in the Relaxation-Driven Cyclic Cosmology

Michael Lehmann

mi.lehmann@gmx.de

April 2026

Abstract

The kinematic Sunyaev–Zel’dovich (kSZ) effect provides a direct probe of cosmic velocity fields and the dynamical evolution of large-scale structure. In the standard cosmological model, the kSZ signal arises from the Doppler shift of CMB photons scattered by free electrons moving with bulk velocities. In the Relaxation-Driven Cyclic Cosmology (RDCC), the scale-dependent evolution of the gravitational potential modifies the growth of velocity fields, the coherence of bulk flows, and the coupling between baryons and dark matter. This Companion develops a continuous description of the kSZ effect in RDCC, deriving how the relaxation scale influences the velocity divergence, the pairwise momentum field, and the angular power spectrum of the kSZ signal. The resulting framework provides a new observational window into the dynamical structure of the RDCC Universe.

1 Introduction

The kinematic Sunyaev–Zel’dovich effect is a Doppler shift imprinted on the cosmic microwave background by free electrons moving with bulk velocities. Unlike the thermal SZ effect, which probes the thermal energy of the intracluster medium, the kSZ effect is sensitive to the momentum field of baryons and thus to the dynamical evolution of large-scale structure.

In the standard cosmological model, the velocity field is governed by the linear growth rate and the continuity equation. In RDCC, the gravitational potential evolves differently. The relaxation mechanism suppresses the decay of small-scale modes and enhances the coherence of large-scale modes. This modifies the velocity divergence, the amplitude of bulk flows, and the coupling between baryons and dark matter. The kSZ effect becomes a direct tracer of the relaxation scale k_{rel} and the temporal structure of the gravitational field.

2 Velocity Fields in RDCC

The cosmic velocity field is governed by the Euler equation,

$$\frac{\partial \mathbf{v}}{\partial a} + \frac{1}{a}(\mathbf{v} \cdot \nabla)\mathbf{v} = -\frac{1}{aH}\nabla\Phi_{\text{RDCC}}. \quad (1)$$

In RDCC, the gravitational potential takes the form

$$\Phi_{\text{RDCC}}(k, a) = \Phi(k, a) \left[1 - \chi_{\Phi} \left(\frac{k}{k_{\text{rel}}} \right)^{\alpha} \right]. \quad (2)$$

The velocity divergence,

$$\theta = \nabla \cdot \mathbf{v}, \quad (3)$$

is therefore modified by the relaxation mechanism. Small-scale modes contribute less to the divergence, while large-scale modes retain their coherence. The velocity field becomes smoother and more coherent on scales near k_{rel} .

3 The RDCC kSZ Signal

The kSZ temperature shift is given by

$$\frac{\Delta T_{\text{kSZ}}}{T_\gamma} = -\sigma_T \int n_e(\mathbf{x}) \frac{\mathbf{v} \cdot \hat{\mathbf{n}}}{c} dl, \quad (4)$$

where $\hat{\mathbf{n}}$ is the line-of-sight direction.

In RDCC, both the electron density and the velocity field carry the imprint of the relaxation scale. The kSZ signal becomes

$$\Delta T_{\text{kSZ,RDCC}} = \Delta T_{\text{kSZ}} \left[1 - \chi_{\text{kSZ}} \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (5)$$

This modification suppresses small-scale kSZ fluctuations and enhances large-scale coherence.

4 Pairwise Momentum and Bulk Flows

The pairwise momentum field,

$$p_{12}(r) = \langle [1 + \delta_e(\mathbf{x})] \mathbf{v}(\mathbf{x}) \cdot \hat{\mathbf{r}} \rangle, \quad (6)$$

is a key observable for the kSZ effect.

In RDCC, the suppression of small-scale potential evolution reduces the amplitude of small-scale pairwise velocities, while the enhanced coherence of large-scale modes increases the amplitude of bulk flows. The pairwise momentum becomes

$$p_{12,\text{RDCC}}(r) = p_{12}(r) \left[1 - \chi_p \left(\frac{1}{k_{\text{rel}} r} \right)^\alpha \right]. \quad (7)$$

This produces a characteristic scale-dependent modulation that reflects the relaxation scale.

5 The kSZ Power Spectrum

The angular power spectrum of the kSZ effect is given by

$$C_\ell^{\text{kSZ}} = \int dk P_q(k) j_\ell^2(kr), \quad (8)$$

where $P_q(k)$ is the power spectrum of the momentum field.

In RDCC, the momentum power spectrum becomes

$$P_{q,\text{RDCC}}(k) = P_q(k) \left[1 - \chi_q \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (9)$$

This modification suppresses small-scale kSZ power and enhances large-scale coherence, producing a distinctive RDCC signature in the kSZ power spectrum.

6 Conclusion

The kinematic Sunyaev–Zel’dovich effect in the Relaxation-Driven Cyclic Cosmology reflects the scale-dependent evolution of the gravitational potential. The relaxation mechanism modifies the velocity divergence, the amplitude of bulk flows, and the coupling between baryons and dark matter. These effects produce distinctive signatures in the kSZ temperature shift, the pairwise momentum field, and the kSZ power spectrum. The present Companion establishes the theoretical foundation for future studies of cosmic velocity fields, baryon dynamics, and the interplay between matter and radiation in RDCC.

References

- [1] Lehmann, M. (2026). Relaxation-Driven Cyclic Cosmology (RDCC): A Minimal CPT-Symmetric Framework with a Single Parameter. Flagship Paper. Zenodo: <https://zenodo.org/records/19744958> (v43.0 or later).
- [2] The complete RDCC companion series (I–LVII) is available at the same Zenodo record above.
- [3] Companion XVI: Boltzmann Code and Modified Hierarchy.
- [4] Companion XXXI: RDCC Halo Model and Non-Linear Structure.
- [5] Companion XXXII–XXXIX: Galaxy–Halo Connection, Cosmic Web, Lensing, RSD, ISW, tSZ, Rotation Curves, CGM.
- [6] Planck Collaboration (2020), *Astron. Astrophys.* 641, A6.
- [7] DESI Collaboration (2024/2025), arXiv: [aktuelle $f\sigma_8$ /BAO-Paper].
- [8] Maiolino et al. (2024), *Nature* (JWST high- z galaxies/quasars).
- [9] Kaiser, N. (1987), *Mon. Not. R. Astron. Soc.* 227, 1 (RSD).
- [10] Bartelmann, M. & Schneider, P. (2001), *Phys. Rep.* 340, 291 (Weak Lensing Review).
- [11] Hu, W. & Okamoto, T. (2002), *Astrophys. J.* 574, 566 (CMB Lensing).
- [12] Lewis, A. & Challinor, A. (2006), *Phys. Rep.* 429, 1 (CMB Lensing Review).
- [13] Navarro, J. F., Frenk, C. S. & White, S. D. M. (1997), *Astrophys. J.* 490, 493 (Halo Profiles).